Micro-ROS Multi-Board Control for a Robotic Leg

Authors: D. F. Gomes, et al. (2026)

ABSTRACT

This paper presents a distributed, real-time control system for a 3D-printed robotic leg. The system uses a modular, two-microcontroller architecture to partition joint control, reducing wiring complexity and improving modularity. It leverages micro-ROS to integrate these embedded controllers into a ROS 2 ecosystem as nodes communicating via the Micro-XRCE-DDS protocol. Real-time control is achieved using FreeRTOS, maintaining a 1 kHz motor-control loop while handling micro-ROS communication in remaining CPU windows to ensure deterministic timing.

RELEVANCE TO THE EB15 ROBOTIC ARM PROJECT

Directly supports the distributed control architecture design of the EB15 robotic arm, validating the partitioning of joint control between master (ESP32 S3) and slave (Arduino Uno) units using modular microcontrollers.

TECHNICAL ANALYSIS & SYSTEM INTEGRATION

This academic research reference documents crucial design paradigms directly relevant to the development of the EB15 six-degrees-of-freedom robotic manipulator. Under the distributed control framework designed for this thesis (featuring the ESP32 S3 as master and the Arduino Uno as slave), the mathematical formulations, communication latencies, and performance profiles analyzed by D. F. Gomes provide a benchmark for physical and algorithmic modeling. Specifically, this reference establishes solid validation criteria for timing constraints, closed-loop feedback via absolute magnetic encoders (AS5600), and vibration damping.